

Maximal Subcodes and Minimal Supercodes of Quasi-Cyclic Codes

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Quasi-cyclic Codes

Let m and ℓ be positive integer with $\gcd(m, q) = 1$. A linear code $\mathcal{C} \subseteq \mathbb{F}_q^{m\ell}$ is called a quasi-cyclic (QC) code of index ℓ if it is invariant under the cyclic shift of codewords by ℓ positions and ℓ is the least positive integer with this property. In particular, if $\ell = 1$, then $\mathcal{C}$ is a cyclic code.

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If we view a codeword $\mathbf{c} \in \mathcal{C}$ as an $m \times \ell$ array

$$\mathbf{c} = \begin{pmatrix} c_{00} & \cdots & c_{0,\ell-1} \\ \vdots & & \vdots \\ c_{m-1,0} & \cdots & c_{m-1,\ell-1} \end{pmatrix},$$

then being invariant under shift by ℓ positions in $\mathbb{F}_q^{m\ell}$ corresponds to being closed under row shift in $\mathbb{F}_q^{m \times \ell}$.

Quasi-cyclic Codes

We set $R = \mathbb{F}_q[x]/\langle x^m - 1 \rangle$ and consider

$$\begin{array}{ccc}
 \Phi : & \mathbb{F}_q^{m\ell} & \longrightarrow R^\ell \\
 \\
 \mathbf{c} = \begin{pmatrix} c_{00} & \dots & c_{0,\ell-1} \\ \vdots & & \vdots \\ c_{m-1,0} & \dots & c_{m-1,\ell-1} \end{pmatrix} & \longmapsto & \begin{array}{c} \mathbf{c}(x) \\ \parallel \\ (c_0(x), \dots, c_{\ell-1}(x)) \end{array} \\
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 \end{array}$$

Observe that the row shift invariance in $\mathbb{F}_q^{m \times \ell}$ corresponds to being closed under componentwise multiplication by x in R^ℓ . Therefore, the map Φ above yields an R -module isomorphism and any QC code $\mathcal{C} \subseteq \mathbb{F}_q^{m\ell} \simeq \mathbb{F}_q^{m \times \ell}$ of index ℓ can be viewed as an R -submodule of R^ℓ .

CRT Decomposition

We assume that $x^m - 1$ factors into irreducible polynomials in $\mathbb{F}_q[x]$ as

$$x^m - 1 = f_1(x)f_2(x) \cdots f_s(x).$$

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$$R \cong \bigoplus_{i=1}^s \mathbb{F}_q[x] / \langle f_i(x) \rangle.$$

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Let ξ be a primitive m^{th} root of unity. Then, the roots of $x^m - 1$ are $1 = \xi^0, \xi, \dots, \xi^{m-1}$. Let $\mathbb{F}$ be the smallest extension of $\mathbb{F}_q$ that contains ξ (equivalently, $\mathbb{F}$ is the splitting field of $x^m - 1$).

CRT Decomposition

For each $i \in \{1, 2, \dots, s\}$, let u_i be the smallest nonnegative integer such that $f_i(\xi^{u_i}) = 0$. Since $f_i(x)$'s are irreducible, the direct summands can be viewed as field extensions of $\mathbb{F}_q$, obtained by adjoining the element ξ^{u_i} .

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If we set $\mathbb{E}_i := \mathbb{F}_q(\xi^{u_i}) \cong \mathbb{F}_q[x]/\langle f_i(x) \rangle$, for each $1 \leq i \leq s$, then $\mathbb{E}_i$ is an intermediate field between $\mathbb{F}$ and $\mathbb{F}_q$ such that $[\mathbb{E}_i : \mathbb{F}_q] = \deg(f_i)$.

CRT Decomposition

We have

$$\begin{aligned} R &\simeq \mathbb{E}_1 \oplus \cdots \oplus \mathbb{E}_s \\ a(x) &\mapsto (a(\xi^{u_1}), \dots, a(\xi^{u_s})). \end{aligned}$$

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This implies that

$$\begin{aligned} R^\ell &\simeq \mathbb{E}_1^\ell \oplus \cdots \oplus \mathbb{E}_s^\ell \\ \mathbf{a}(x) &\mapsto (\mathbf{a}(\xi^{u_1}), \dots, \mathbf{a}(\xi^{u_s})), \end{aligned}$$

where $\mathbf{a}(\delta)$ denotes the componentwise evaluation at $\delta \in \mathbb{F}$, for any $\mathbf{a}(x) = (a_0(x), \dots, a_{\ell-1}(x)) \in R^\ell$.

Constituents

Hence, a QC code $\mathcal{C} \subseteq R^\ell$ decomposes as

$$\mathcal{C} \simeq \mathcal{C}_1 \oplus \cdots \oplus \mathcal{C}_s,$$

where $\mathcal{C}_i$ is a linear code in $\mathbb{F}_q^\ell$, for each i . These linear codes over various extensions of $\mathbb{F}_q$ are called the *constituents* of $\mathcal{C}$.

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Let $\mathcal{C} \subseteq R^\ell$ be generated as an R -module by

$$\left\{ (a_{1,0}(x), \dots, a_{1,\ell-1}(x)), \dots, (a_{r,0}(x), \dots, a_{r,\ell-1}(x)) \right\}.$$

Then, for $1 \leq i \leq s$, we have

$$\mathcal{C}_i = \text{span}_{\mathbb{F}_i} \left\{ (a_{b,0}(\xi^{u_i}), \dots, a_{b,\ell-1}(\xi^{u_i})) : 1 \leq b \leq r \right\}.$$

Constituents

Note that each field $\mathbb{E}_i$ is isomorphic to a minimal cyclic code of length m over $\mathbb{F}_q$; namely, the cyclic code in $\mathbb{F}_q^m$ with the irreducible check polynomial $f_i(x)$. If we denote by θ_i the generating primitive idempotent for the minimal cyclic code $\langle \theta_i \rangle$ in consideration, then the isomorphism is given by the maps

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$$\begin{array}{ll} \varphi_i : \langle \theta_i \rangle & \longrightarrow \mathbb{E}_i \\ a(x) & \longmapsto a(\xi^{u_i}) \end{array} \qquad \begin{array}{ll} \psi_i : \mathbb{E}_i & \longrightarrow \langle \theta_i \rangle \\ \delta & \longmapsto \sum_{k=0}^{m-1} a_k x^k, \end{array}$$

where

$$a_k = \frac{1}{m} \operatorname{Tr}_{\mathbb{E}_i/\mathbb{F}_q}(\delta \xi^{-ku_i}).$$

Concatenation

Theorem 1

- i. Let $\mathcal{C}$ be an R -submodule of R^ℓ (i.e., a q -ary QC code).
Then, for some subset $\mathcal{I}$ of $\{1, \dots, s\}$, there exist linear codes $\mathfrak{C}_i \subseteq \mathbb{E}_i^\ell$ such that

$$\mathcal{C} = \bigoplus_{i \in \mathcal{I}} \langle \theta_i \rangle \square \mathfrak{C}_i. \quad (1)$$

- ii. Conversely, let $\mathfrak{C}_i$ be a linear code over $\mathbb{E}_i$ of length ℓ , for each $i \in \mathcal{I} \subseteq \{1, \dots, s\}$. Then,

$$\mathcal{C} = \bigoplus_{i \in \mathcal{I}} \langle \theta_i \rangle \square \mathfrak{C}_i$$

is a q -ary QC code of length $m\ell$ and index ℓ .

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Moreover, each constituent $\mathcal{C}_i$ is equal to the outer code $\mathfrak{C}_i$ in the concatenated structure, for each i .

Subcodes-Supercodes

Given a QC code $\mathcal{C}$ of length $m\ell$ and index ℓ with the concatenated structure given in Theorem 1 above, let $\mathcal{C}'$ be another QC code of the same length and index such that

$$\mathcal{C}' = \bigoplus_{i \in \mathcal{I}'} \langle \theta_i \rangle \square \mathcal{C}'_i, \quad (2)$$

where $\mathcal{I}' \subseteq \{1, \dots, s\}$. Then, the following is easy to observe.

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Proposition 1

Given two QC codes $\mathcal{C}$ and $\mathcal{C}'$ of length $m\ell$ and index ℓ with the concatenated structures as in (1) and (2), then $\mathcal{C}'$ is a QC-subcode of $\mathcal{C}$ if and only if $\mathcal{I}' \subseteq \mathcal{I}$ and $\mathcal{C}'_i \subseteq \mathcal{C}_i$, for all $i \in \mathcal{I}'$. Conversely, $\mathcal{C}'$ is a QC-supercode of $\mathcal{C}$ if and only if $\mathcal{I} \subseteq \mathcal{I}'$ and $\mathcal{C}_i \subseteq \mathcal{C}'_i$, for all $i \in \mathcal{I}$.

Definition 2

Assume that $\mathcal{C}$ and $\mathcal{C}'$ are two QC codes of length $m\ell$ and index ℓ with the concatenated structures as in (1) and (2). Allowing $\dim(\mathcal{C}_i) = 0$, we can w.l.o.g. assume $\mathcal{I} = \mathcal{I}'$. Then $\mathcal{C}'$ is a maximal QC-subcode (minimal QC-supercode) of $\mathcal{C}$ if $\mathcal{C}'_i \subsetneq \mathcal{C}_i$ and $\dim(\mathcal{C}'_i) = \dim(\mathcal{C}_i) - 1$ ($\mathcal{C}_i \subsetneq \mathcal{C}'_i$ and $\dim(\mathcal{C}'_i) = \dim(\mathcal{C}_i) + 1$), for some $i \in \mathcal{I}$, and $\mathcal{C}_j = \mathcal{C}'_j$, for all $j \in \mathcal{I}, j \neq i$.

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Assuming the notation above, we set

$e_i := [\mathbb{E}_i : \mathbb{F}_q] = \deg(f_i(x)) = \dim(\langle \theta_i \rangle)$. Then,

$$m = \sum_{i=1}^s e_i \quad \text{and} \quad \dim(\mathcal{C}) = \sum_{i \in \mathcal{I}} e_i \cdot \dim(\mathcal{C}_i). \quad (3)$$

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Comparing the dimension of $\mathcal{C}$ with the dimension of its maximal QC-subcode or minimal QC-supercode $\mathcal{C}'$, we obtain the following.

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Proposition 2

Given two QC codes $\mathcal{C}$ and $\mathcal{C}'$ of length $m\ell$ and index ℓ such that $\mathcal{C}'$ is a maximal QC-subcode (minimal QC-supercode) of $\mathcal{C}$, we have $\dim(\mathcal{C}') = \dim(\mathcal{C}) - e_i$ ($\dim(\mathcal{C}') = \dim(\mathcal{C}) + e_i$), for some $i \in \mathcal{I}$.

Repeated-root Case

Now let $\gcd(q, m) = p^a$ with $a > 0$. Then, $m = p^a m'$, where $\gcd(m', q) = 1$. The polynomial $x^m - 1$ has the factorization

$$x^m - 1 = (x^{m'} - 1)^{p^a}.$$

Similar as before, we consider the factorization

$$x^{m'} - 1 = f_1(x)f_2(x)\dots f_s(x).$$

By the CRT, we have the following ring isomorphism:

$$\mathbb{F}_q[x]/\langle x^m - 1 \rangle = \mathbb{F}_q[x]/\langle (x^{m'} - 1)^{p^a} \rangle \cong \bigoplus_{i=1}^s \mathbb{F}_q[x]/\langle f_i(x)^{p^a} \rangle.$$

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As above, we let $\mathbb{E}_i = \mathbb{F}_q[x]/\langle f_i(x) \rangle$ and we obtain an isomorphism between $\mathbb{F}_q[x]/\langle f_i(x)^{p^a} \rangle$ and $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$ given by

$$\Phi_i: x \mapsto \xi^{u_i}(1 - v),$$

where ξ is a primitive m' -th root of unity and $f_i(\xi^{u_i}) = 0$.

Repeated-root Case

Similarly, a QC code $\mathcal{C}$ of length $m\ell$ and index ℓ over $\mathbb{F}_q$ with $\gcd(q, m) = p^a$ decomposes as

$$\mathcal{C} \cong \mathcal{C}_1 \oplus \cdots \oplus \mathcal{C}_s,$$

where $\mathcal{C}_i$ is a linear code of length ℓ over S_i , for each $i = 1, \dots, s$.

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Lemma 1

The inverse of the map $\Phi_i: x \mapsto \xi^{u_i}(1 - v)$ is given by

$$\Psi_i: \begin{cases} v \mapsto 1 - x^{1-p^a\mu} \\ \xi^{u_i} \mapsto x^{p^a\mu}, \end{cases}$$

where $\xi^{u_i p^a \mu} = \xi^{u_i}$, i.e., μ is the inverse of p^a modulo m' .

Repeated-root Case

Applying Ψ_i componentwise to the constituents C_i , we get codes over $\mathbb{F}_q[x]/\langle f_i(x)^{p^a} \rangle$. In order to invert the CRT decomposition, the image of Ψ_i is multiplied by $g_i(x)h_i(x)$ where $g_i(x) := \prod_{i \neq j} f_j(x)^{p^a}$ and $h_i(x)$ is the inverse of $g_i(x)$ modulo $f_i(x)^{p^a}$.

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As $\mathbb{E}_i$ -vector space, we can identify S_i with $\mathbb{E}_i^{p^a}$ using

$$b(v) = \sum_{j=0}^{p^a-1} b_j v^j \mapsto (b_0, b_1, \dots, b_{p^a-1}).$$

This can be extended to an $\mathbb{E}_i$ -isomorphism between S_i^ℓ and $\mathbb{E}_i^{\ell p^a}$, where we group the coefficients according to the degree of v .

Repeated-root Case

Expanding an S_i -linear code C of length ℓ , we obtain a code $\tilde{C}$ over $\mathbb{E}_i$ of length ℓp^a . Multiplication by v over S_i corresponds to shifting the expanded vector by ℓ positions to the right without wrap-around, i.e., leading positions are filled with zeros. A generator matrix for the code over $\mathbb{E}_i$ has the form

$$\begin{pmatrix} G_{1,1} & G_{1,2} & G_{1,3} & \cdots & G_{1,p^a} \\ 0 & G_{1,1} & G_{1,2} & \cdots & G_{1,p^a-1} \\ 0 & G_{2,2} & G_{2,3} & \cdots & G_{2,p^a} \\ \hline 0 & 0 & G_{1,1} & \cdots & G_{1,p^a-2} \\ 0 & 0 & G_{2,2} & \cdots & G_{2,p^a-1} \\ 0 & 0 & G_{3,3} & \cdots & G_{3,p^a} \\ \hline \vdots & \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & 0 & \cdots & G_{1,1} \\ 0 & 0 & 0 & \cdots & G_{2,2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & G_{p^a,p^a} \end{pmatrix}.$$

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Proposition 3

Let $C \subset S_i^\ell$ be a linear code over $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$, i. e., an S_i -module. For a vector $\mathbf{y} \in S_i^\ell$ with $\mathbf{y} \notin C$, the code $C' = \langle C, \mathbf{y} \rangle_{S_i}$ is a minimal supercode of C if and only if $v\mathbf{y} \in C$.

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Corollary 3

Let $C \subset S_i^\ell$ be a linear code over $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$, i. e., an S_i -module. Assume that $C = C' \oplus \langle \mathbf{y} \rangle$. The code C' is a maximal subcode of C if and only if $v\mathbf{y} \in C'$ and $\mathbf{y} \notin \text{shift}_\ell(C)$.

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→ 174 new codes and 24 more from Construction X and variants

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